

CLASS XII · LECTURE 12

Database Fundamentals

Both options · Two periods · Asked inside Paper II whichever language you chose

By the end of this lecture you can

- Distinguish a flat file from a database.
- Define a table, a record and a field, and give the other names the paper uses.
- List the components of a DBMS and three duties of a DBA.
- State advantages that match the multiple-choice wording.

From a pile of files to a database

A **flat file** is one file of records, often repeated in another file. A college might keep a class list in one sheet and a fee list in another, with each student's name typed twice. Change a name in one place and the other place is wrong. A **database** stores the data once, in related tables, and many programs can use it. A **database management system** (DBMS) is the software that creates, protects and queries that database. Microsoft Access is the DBMS this paper expects you to know. The person who writes application programs is a programmer. The person who looks after the database is the DBA. The person who only uses the finished screens is an end user.

DEFINITION Table, record, field

A table is the basic object: rows and columns about one kind of thing, such as Student. A row is a record (also called a tuple). A column is a field (also called an attribute). Roll number, name and class are fields. Ali's whole line is one record.

Word in one book	Word in another	In a sentence
Row	Record or tuple	One student's data
Column	Field or attribute	The Name column
Table	Relation	The Student table
Table structure	Schema	The field names and types, not the rows of data

Components of a DBMS

Five components answer "name the major components of a DBMS". Write them as a list with a few words each.

Component	What it is
Hardware	The computer, disks and network the database runs on
Software	The DBMS itself, such as MS Access, plus the operating system
Data	The facts stored in the tables. This is the reason the system exists
Procedures	The rules: who may enter marks, how often a backup is taken
Users and the data language	DBA, programmers and end users. SQL is the language they use to ask questions

Why a database is worth the trouble

Learn five advantages in verbs. **Redundancy is reduced** because a fact is stored once. **Consistency** follows: there is one name to update. **Sharing** means several programs, and several users, can use the same data. **Integrity** means the data obeys rules, such as “marks are between 0 and 100”. **Security** means accounts and passwords decide who may see or change a table. Backup and recovery are a sixth, and a good one. The multiple-choice advantage that matches this list is: data is integrated and can be accessed by multiple programs. “Redundancy increases” and “integrity decreases” are the wrong way round.

ON THE BOARD Three duties of a DBA

The database administrator designs the tables, controls who may use them, and takes backups so the data can be restored. Monitoring performance and keeping the data consistent are fair extra duties. Do not say the DBA's job is to type every student's marks. That is a data-entry user.

Applications you can quote: a college admission system, a bank's account records, a hospital's patients, a library's loans, a shop's stock. One example beside the definition is enough in Section B.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. The primary object in a database is the:

- A) Report
- B) Query
- C) Table
- D) Form

2. A row in a table is a:

- A) Field
- B) Record
- C) Query
- D) Form

3. An advantage of the database approach is that:

- A) Data redundancy increases
- B) Data is integrated and can be shared by programs
- C) Integrity must decrease
- D) Each program keeps its own copy on purpose

Short answers

S1. Name the major components of a DBMS.

S2. Who is a DBA? Write three responsibilities.

S3. Define table, query and report.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. C. Reports and queries are built on tables. The data lives in the table.

MCQ 2. B. A column is a field. A row is a record.

MCQ 3. B. The point of a DBMS is one integrated store used by many programs.

S1. Hardware (the machine and disks), software (the DBMS, such as MS Access), data (the contents of the tables), procedures (backup and access rules), and people together with a data language such as SQL. Users include the DBA, programmers and end users.

S2. A database administrator is the person responsible for the database. Three duties are: designing the tables and relationships, controlling security so only the right people can change data, and taking backups and restoring them when needed.

S3. A table stores the data in rows and columns. A query asks a question of the tables, for example which students scored 80 or more, and can also change data. A report lays the data out for printing. A form, if you have a fourth line, is the screen used to enter and view records.